

|                                 |                                    | BCIM Engineering Pvt. Ltd.                                                       |                                 |                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                              |          |       |         |
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| INSPECTION AND TEST PLAN        |                                    |                                                                                  |                                 |                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                 |          |       |         |
| PROJECT                         |                                    | Lanco Hills Residential Project                                                  |                                 |                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                 |          |       |         |
| BUILDINGS                       |                                    | LH10                                                                             |                                 |                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                 |          |       |         |
| PART - A - MATERIAL & EQUIPMENT |                                    |                                                                                  |                                 |                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                 |          |       |         |
| Sl.No.                          | Inspection and Testing Description | Controlling Specification                                                        | Verifying Document              | Frequency of Inspec./Testing                                                                                                                                      | Acceptance Criteria                                                                                                                                                                                                                                                                                                                                                                                | Responsibility                                                                                                                                                                                  |          |       | Remarks |
|                                 |                                    |                                                                                  |                                 |                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                    | Vendor                                                                                                                                                                                          | Client   |       |         |
|                                 |                                    |                                                                                  |                                 |                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                 | Projects | QA/QC |         |
| 1                               | CEMENT                             |                                                                                  |                                 |                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                 |          |       |         |
|                                 | Visual Inspection ( Bag Cement)    | Grade & Week No.                                                                 | IMIR                            | Random                                                                                                                                                            | Purchase order Specification                                                                                                                                                                                                                                                                                                                                                                       | C                                                                                                                                                                                               | C        | C     |         |
|                                 | Compressive Strength               | IS:269 : 2015 & 4031(Part-6)<br>(43 & 53 Grade Cement)<br>IS : 3535 Table 2      | MTC /<br>Third Party Lab Report | MTC - Bulker / Bag Cement - Every Lot/Batch<br>Third Party Lab Report - Bag Cement - Once in a month,<br>Third Party Lab Report - Bulker Cement - Once in a month | <u>Age</u><br>a) 72 ± 1 h<br>b) 168 ± 2 h<br>c) 672 ± 4 h                                                                                                                                                                                                                                                                                                                                          | <u>Comp. Strength in Mpa.</u><br>43 Grade OPC :<br>a. Min. 23 , b. Min. 33, c. Min. 43 & Max. 53<br>53 Grade OPC:<br>a. Min.27, b. Min.37 c. Min. 53                                            | R        | R     | R       |
|                                 | Initial and Final Setting Time     | IS:269 : 2015 & 4031(Part-6)<br>(43 & 53 Grade Cement)<br>IS : 3535 Table 2      | MTC /<br>Third Party Lab Report |                                                                                                                                                                   | Setting time: IS 4031 (Part 5)<br>a) Initial<br>b) Final                                                                                                                                                                                                                                                                                                                                           | a) Min 30 mins<br>b) Max 600 mins                                                                                                                                                               | R        | R     | R       |
|                                 | Fineness                           | IS:269 : 2015 & 4031(Part-6)<br>(43 & 53 Grade Cement)<br>IS : 3535 Table 2      | MTC /<br>Third Party Lab Report |                                                                                                                                                                   | Fineness, m2/kg,                                                                                                                                                                                                                                                                                                                                                                                   | Min 225                                                                                                                                                                                         | R        | R     | R       |
|                                 | Soundness                          | IS:269 : 2015 & 4031(Part-6),<br>(43 & 53 Grade Cement)<br>IS : 3535 Table 2     | MTC /<br>Third Party Lab Report | MTC - Bulker / Bag Cement - Every Lot/Batch<br>Third Party Lab Report - Bag Cement - Once in a month<br>Third Party Lab Report - Bulker Cement - Once in a month  | Soundness: IS 4031 (Part 3)<br>a) By Le Chatelier method, mm,<br>b) By autoclave test method, percent                                                                                                                                                                                                                                                                                              | a) Max 10<br>b) Max 0.8                                                                                                                                                                         | R        | R     | R       |
|                                 | Chemical Properties                | IS:269 : 2015 & 4031(Part-6).4032<br>(43 & 53 Grade Cement)<br>IS : 3535 Table 2 | MTC /<br>Third Party Lab Report | MTC - Bulker / Bag Cement - Every Lot/Batch<br>Third Party Lab Report - Bag Cement - Once in a month<br>Third Party Lab Report - Bulker Cement - Once in a month  | Ratio of percentage of lime to percentages of silica,<br>alumina and iron oxide,<br><br>Ratio of percentage of alumina to that of iron oxide<br><br>Insoluble residue, percent by mass<br><br>Magnesia, percent by mass<br><br>Total sulphur content calculated as sulphuric<br>anhydride (SO3), percent by mass<br><br>Loss on ignition, percent by mass<br><br>Chloride content, percent by mass | 43 Grade OPC: 0.66 - 1.02<br>53 Grade OPC : 0.80 - 1.02<br><br>Min 0.66<br><br>Max 5.00<br><br>Max 6.00<br><br>Max 3.5<br><br>43 Grade OPC : Max. 5.0<br>53 Grade OPC : Max. 4.0<br><br>Max 0.1 | R        | R     | R       |

|                                 |                                                                                                                                                                                                                                                                                                                                                  | BCIM Engineering Pvt. Ltd.      |                              |                                                                                                                    |                                                                  |                                                 |  |          |       |         |
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| INSPECTION AND TEST PLAN        |                                                                                                                                                                                                                                                                                                                                                  |                                 |                              |                                                                                                                    |                                                                  |                                                 |                                                                                     |          |       |         |
| PROJECT                         |                                                                                                                                                                                                                                                                                                                                                  | Lanco Hills Residential Project |                              |                                                                                                                    |                                                                  |                                                 |                                                                                     |          |       |         |
| BUILDINGS                       |                                                                                                                                                                                                                                                                                                                                                  | LH10                            |                              |                                                                                                                    |                                                                  |                                                 |                                                                                     |          |       |         |
| PART - A - MATERIAL & EQUIPMENT |                                                                                                                                                                                                                                                                                                                                                  |                                 |                              |                                                                                                                    |                                                                  |                                                 |                                                                                     |          |       |         |
| Sl.No.                          | Inspection and Testing Description                                                                                                                                                                                                                                                                                                               | Controlling Specification       | Verifying Document           | Frequency of Inspec./Testing                                                                                       | Acceptance Criteria                                              |                                                 | Responsibility                                                                      |          |       | Remarks |
|                                 |                                                                                                                                                                                                                                                                                                                                                  |                                 |                              |                                                                                                                    |                                                                  |                                                 | Vendor                                                                              | Client   |       |         |
|                                 |                                                                                                                                                                                                                                                                                                                                                  |                                 |                              |                                                                                                                    |                                                                  |                                                 |                                                                                     | Projects | QA/QC |         |
| 2                               | REINFORCEMENT STEEL                                                                                                                                                                                                                                                                                                                              |                                 |                              |                                                                                                                    |                                                                  |                                                 |                                                                                     |          |       |         |
|                                 | Visual Inspection                                                                                                                                                                                                                                                                                                                                | Purchase order                  | IMIR                         | Every Consignment                                                                                                  | MTC                                                              |                                                 | C                                                                                   | C        | C     |         |
|                                 | Unit Weight Per mtr at Site                                                                                                                                                                                                                                                                                                                      | IS:1786-2008                    | MTC / Third Party Lab Report | MTC - Every lot /batch<br>Third Party Lab Report- One test / 50MT or Part thereof per Dia per consignment per day  | Nominal Size mm                                                  | Batch Sample Individual Sample For coils        | R                                                                                   | R        | R     |         |
|                                 |                                                                                                                                                                                                                                                                                                                                                  |                                 |                              |                                                                                                                    | Up to and including 10                                           | +/- 7 -8 +/-8                                   |                                                                                     |          |       |         |
|                                 |                                                                                                                                                                                                                                                                                                                                                  |                                 |                              |                                                                                                                    | Over 10 and Up to and including 16 mm                            | +/- 5 -6 +/-6                                   |                                                                                     |          |       |         |
|                                 |                                                                                                                                                                                                                                                                                                                                                  |                                 |                              |                                                                                                                    | Over 16                                                          | +/- 3 -4 +/-4                                   |                                                                                     |          |       |         |
|                                 | 0.2 percent proof stress / yield stress, Min N/Sqmm                                                                                                                                                                                                                                                                                              | IS:1786-2008                    | MTC / Third Party Lab Report | MTC - Every lot /batch<br>Third Party Lab Report - One test / 50MT or Part thereof per Dia per consignment per day | Fe 500 - 500<br>Fe 500 D - 500<br>Fe 550 - 550<br>Fe 550 D - 550 |                                                 | R                                                                                   | R        | R     |         |
|                                 | Elongation, percent, Min.                                                                                                                                                                                                                                                                                                                        | IS:1786-2008                    | MTC / Third Party Lab Report | MTC - Every lot /batch<br>Third Party Lab Report - One test / 50MT or Part thereof per Dia per consignment per day | Fe 500 - 12<br>Fe 500 D - 16<br>Fe 550 - 10<br>Fe 550 D - 14.5   |                                                 | R                                                                                   | R        | R     |         |
|                                 | Bend and Rebend                                                                                                                                                                                                                                                                                                                                  | IS:1786-2008                    | MTC / Third Party Lab Report | MTC - Every lot /batch<br>Third Party Lab Report - One test / 50MT or Part thereof per Dia per consignment per day | Bend - Passes If no Cracks or Rupture Visible                    | Rebend - Passes If no Cracks or Rupture Visible | R                                                                                   | R        | R     |         |
|                                 | Tensile Strength Min                                                                                                                                                                                                                                                                                                                             | IS:1786-2008                    | MTC / Third Party Lab Report | MTC - Every lot /batch<br>Third Party Lab Report - One test / 50MT or Part thereof per Dia per consignment per day | Fe 500                                                           | ≥ 1.08, but TS not less than 545 N/sqmm         | R                                                                                   | R        | R     |         |
|                                 |                                                                                                                                                                                                                                                                                                                                                  |                                 |                              |                                                                                                                    | Fe 500 D                                                         | ≥ 1.10, but TS not less than 565 N/sqmm         |                                                                                     |          |       |         |
|                                 |                                                                                                                                                                                                                                                                                                                                                  |                                 |                              |                                                                                                                    | Fe 550                                                           | ≥ 1.06, but TS not less than 585 N/sqmm         |                                                                                     |          |       |         |
|                                 |                                                                                                                                                                                                                                                                                                                                                  |                                 |                              |                                                                                                                    | Fe 550 D                                                         | ≥ 1.08, but TS not less than 600 N/sqmm         |                                                                                     |          |       |         |
|                                 | Chemical Properties                                                                                                                                                                                                                                                                                                                              | IS:1786-2008                    | MTC / Third Party Lab Report | MTC - Every lot /batch<br>Third Party Lab Report - One test / 50MT or Part thereof per Dia per consignment per day | <div>Fe 500 Fe 500 D Fe 550 F550D</div>                          |                                                 | R                                                                                   | R        | R     |         |
|                                 | Note :<br>Tolerances as per IS 1786 -2008 Clause 4.2.1<br>For guaranteed weldability carbon equivalent for all grades of steel shall not be more than 0.53 and shall not be more than 0.42, when micro alloys are used<br>When strengthening elements like Nb, V, B, Ti are used individually or in combination, the total shall not exceed 0.30 |                                 |                              |                                                                                                                    | Carbon,% Maximum                                                 | <div>0.300 0.250 0.300 0.250</div>              |                                                                                     |          |       |         |
|                                 |                                                                                                                                                                                                                                                                                                                                                  |                                 |                              |                                                                                                                    | Sulphur,% Maximum                                                | <div>0.055 0.040 0.055 0.040</div>              |                                                                                     |          |       |         |
|                                 |                                                                                                                                                                                                                                                                                                                                                  |                                 |                              |                                                                                                                    | Phosphorus,% Maximum                                             | <div>0.055 0.040 0.055 0.040</div>              |                                                                                     |          |       |         |
|                                 |                                                                                                                                                                                                                                                                                                                                                  |                                 |                              |                                                                                                                    | Sulphur & Phosphorus % Maximum                                   | <div>0.105 0.075 0.100 0.075</div>              |                                                                                     |          |       |         |

|                                                               |                                                                | BCIM Engineering Pvt. Ltd.                                   |                                   |                                                                                                                                                                                    |                                                                                                                         |  |          |       |                                                   |  |
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| INSPECTION AND TEST PLAN                                      |                                                                |                                                              |                                   |                                                                                                                                                                                    |                                                                                                                         |                                                                                     |          |       |                                                   |  |
| PROJECT                                                       |                                                                | Lanco Hills Residential Project                              |                                   |                                                                                                                                                                                    |                                                                                                                         |                                                                                     |          |       |                                                   |  |
| BUILDINGS                                                     |                                                                | LH10                                                         |                                   |                                                                                                                                                                                    |                                                                                                                         |                                                                                     |          |       |                                                   |  |
| PART - A - MATERIAL & EQUIPMENT                               |                                                                |                                                              |                                   |                                                                                                                                                                                    |                                                                                                                         |                                                                                     |          |       |                                                   |  |
| Sl.No.                                                        | Inspection and Testing Description                             | Controlling Specification                                    | Verifying Document                | Frequency of Inspec./Testing                                                                                                                                                       | Acceptance Criteria                                                                                                     | Responsibility                                                                      |          |       | Remarks                                           |  |
|                                                               |                                                                |                                                              |                                   |                                                                                                                                                                                    |                                                                                                                         | Vendor                                                                              | Client   |       |                                                   |  |
|                                                               |                                                                |                                                              |                                   |                                                                                                                                                                                    |                                                                                                                         |                                                                                     | Projects | QA/QC |                                                   |  |
| 3                                                             | BINDING WIRE                                                   |                                                              |                                   |                                                                                                                                                                                    |                                                                                                                         |                                                                                     |          |       |                                                   |  |
|                                                               | Visual Inspection                                              | Purchase order                                               | IMIR                              | Random                                                                                                                                                                             | AS per Relevant Standards                                                                                               | R                                                                                   | R        | R     |                                                   |  |
|                                                               | Mass, Tensile strength, coating thickness, Chemical properties | IS : 280 /Table 2<br>IS : 2629                               | MTC/Third Party Lab Report        | As per Table 3 of IS: 280 -2006 / One test per source                                                                                                                              | AS per Relevant Standards                                                                                               | R                                                                                   | R        | R     |                                                   |  |
| 4                                                             | FINE & COARSE AGGREGATES                                       |                                                              |                                   |                                                                                                                                                                                    |                                                                                                                         |                                                                                     |          |       |                                                   |  |
|                                                               | Visual Inspection                                              | Purchase order                                               | IMIR                              | Random                                                                                                                                                                             | Purchase order Specification                                                                                            | C                                                                                   | R        | R     |                                                   |  |
|                                                               | Specific gravity (FA & CA)                                     | IS:2386(Part-3), 383                                         | Third Party Lab Report            | Third Party Lab Report : One test per source / Year                                                                                                                                |                                                                                                                         | R                                                                                   | R        | R     |                                                   |  |
|                                                               | Bulk Density (FA & CA)                                         | IS:2386(Part-3), 383                                         | Third Party Lab Report            | Third Party Lab Report : One test per source / Year                                                                                                                                |                                                                                                                         | R/C                                                                                 | R        | R     |                                                   |  |
|                                                               | Moisture Content & Batch Correction (FA & CA)                  | Field Test                                                   | Internal Lab Report               | Daily                                                                                                                                                                              |                                                                                                                         | R/C                                                                                 | R        | R     |                                                   |  |
|                                                               | Sieve Analysis/ Gradation (FA & CA)                            | IS:2386(Part-1), 383                                         | Internal / Third Party Lab Report | Third Party Lab Report - One per source / Year<br>Internal Lab Report - One per source / 15days                                                                                    | For Concreting - As per Limits specified in IS 383                                                                      | R/C                                                                                 | R        | R     |                                                   |  |
|                                                               | Flakiness & Elongation Index (CA)                              | IS:2386(Part-1), 383                                         | Internal / Third Party Lab Report | Third Party Lab Report - One per source / Year<br>Internal Lab Report- One per source / 15days                                                                                     | Combined Flakiness & elongation shall be not more than 40%                                                              | R/C                                                                                 | R        | R     |                                                   |  |
|                                                               | Aggregate Impact Value (CA)                                    | IS:2386(Part-4), 383                                         | Internal / Third Party Lab Report | Third Party test - One per source / Year<br>Internal Lab Report - One per source / 90 days                                                                                         | Impact Value :<br>Maximum 30% - concrete for wearing surfaces<br>Maximum 45% -concrete for other than wearing surface   | R/C                                                                                 | R        | R     |                                                   |  |
|                                                               | Alkali Aggregate Reactivity (CA & FA)                          | IS:2386(Part-7), 383                                         | Third Party Lab Report            | Third Party Lab Report - One per source / Year                                                                                                                                     | Crushing Value :<br>Maximum 30% - concrete for wearing surfaces<br>Maximum 30% -concrete for other than wearing surface | R                                                                                   | R        | R     |                                                   |  |
| 5                                                             | WATER                                                          |                                                              |                                   |                                                                                                                                                                                    |                                                                                                                         |                                                                                     |          |       |                                                   |  |
|                                                               | Properties test for Construction Purposes                      | Table:1 of IS:456, 3025<br>IS 4926 : Annexure B, B2 Water(b) | Third Party Lab Report            | A Sample shall be taken annually for routine records however if the chloride content is more than 0.01% the interval testing shall be reduced to 3 months                          | Organic solids                                                                                                          | Max 200 Mg/l                                                                        | R        | R     | R                                                 |  |
| Inorganic Solids                                              |                                                                |                                                              |                                   |                                                                                                                                                                                    | Max 3000 Mg/l                                                                                                           |                                                                                     |          |       |                                                   |  |
| Sulphate as SO <sub>4</sub>                                   |                                                                |                                                              |                                   |                                                                                                                                                                                    | Max 400 Mg/l                                                                                                            |                                                                                     |          |       |                                                   |  |
| Chloride as Cl                                                |                                                                |                                                              |                                   |                                                                                                                                                                                    | Max 500 Mg/l for RCC<br>Max 2000 Mg/l for PCC                                                                           |                                                                                     |          |       |                                                   |  |
| Suspended solids                                              |                                                                |                                                              |                                   |                                                                                                                                                                                    | Max 2000 Mg/l                                                                                                           |                                                                                     |          |       |                                                   |  |
| pH                                                            |                                                                |                                                              |                                   |                                                                                                                                                                                    | Min 6.0                                                                                                                 |                                                                                     |          |       |                                                   |  |
| To neutralize 100 ml of water using Phenolphthalein indicator |                                                                |                                                              |                                   |                                                                                                                                                                                    | 0 - 5.0 ml                                                                                                              |                                                                                     |          |       |                                                   |  |
| To neutralize 100 ml of water using Mixed indicator           |                                                                |                                                              |                                   |                                                                                                                                                                                    | 0 - 25.0 ml                                                                                                             |                                                                                     |          |       |                                                   |  |
| 6                                                             | CONCRETE MIX - TRAILS                                          |                                                              |                                   |                                                                                                                                                                                    |                                                                                                                         |                                                                                     |          |       |                                                   |  |
|                                                               | Workability                                                    | IS:1199 & Approved Concrete Mix design                       | Internal Lab Report               | IS 456 Clause : 9.2.3 - Mix design done earlier not prior to one year may be considered adequate for latter work provided there is no change in source and the quality of material | +/- 25 mm or +/- 1/3 of the specified value                                                                             | C                                                                                   | R/W      | R/W   | This has to be approved by consultants before use |  |
|                                                               | Loss of Workability ( Retention time)                          | IS: 456 Clause 5.5.3 ( To be captured in Mix design)         | Internal Lab Report               |                                                                                                                                                                                    | Actuals (to be designed for 3hrs Retention for External RMC Product or 2hrs Retention for insitu RMC Product)           | C                                                                                   | R/W      | R/W   |                                                   |  |
|                                                               | Compressive Strength                                           | IS:456, 516 & Technical Specifications                       | Internal / Third Party Lab Report |                                                                                                                                                                                    | Min. Designed Strength                                                                                                  | C                                                                                   | C        | R/W   |                                                   |  |
| 7                                                             | CONCRETE                                                       |                                                              |                                   |                                                                                                                                                                                    |                                                                                                                         |                                                                                     |          |       |                                                   |  |

|                                         |                                      | BCIM Engineering Pvt. Ltd.                                                                                               |                                               |                                                                                                                     |                                                                                                           |  |          |       |         |
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| INSPECTION AND TEST PLAN                |                                      |                                                                                                                          |                                               |                                                                                                                     |                                                                                                           |                                                                                     |          |       |         |
| PROJECT                                 |                                      | Lanco Hills Residential Project                                                                                          |                                               |                                                                                                                     |                                                                                                           |                                                                                     |          |       |         |
| BUILDINGS                               |                                      | LH10                                                                                                                     |                                               |                                                                                                                     |                                                                                                           |                                                                                     |          |       |         |
| PART - A - MATERIAL & EQUIPMENT         |                                      |                                                                                                                          |                                               |                                                                                                                     |                                                                                                           |                                                                                     |          |       |         |
| Sl.No.                                  | Inspection and Testing Description   | Controlling Specification                                                                                                | Verifying Document                            | Frequency of Inspec./Testing                                                                                        | Acceptance Criteria                                                                                       | Responsibility                                                                      |          |       | Remarks |
|                                         |                                      |                                                                                                                          |                                               |                                                                                                                     |                                                                                                           | Vendor                                                                              | Client   |       |         |
|                                         |                                      |                                                                                                                          |                                               |                                                                                                                     |                                                                                                           |                                                                                     | Projects | QA/QC |         |
|                                         | Workability of Fresh Concrete(Slump) | IS:1199<br>IS: 456 Clause 5.5.3 ( To be captured in Mix design)<br>IS: 4926 Clause : 10.2.2.1 & Technical Specifications | Batch Sheet                                   | Random<br>(Workability is recorded on the concrete batch sheet)                                                     | +/- 25 mm or +/- 1/3 of the specified value                                                               | C                                                                                   | R/W      | C/R   |         |
|                                         | Compressive strength of concrete     | IS:456, 516 &<br>Cube Sampling & Testing to be done<br>IS:456 - Clause 15.2.2, 516                                       | Internal / Third Party Lab Report             | Internal Lab Report : All Samples<br>Third Party Lab Report -Random                                                 | The average of the 3 consecutive cubes shall be not less than the specified strength ( Table 2 of IS 456) | C /R                                                                                | R/W      | R/W   |         |
|                                         | Standard Deviation                   | IS : 456                                                                                                                 | Internal                                      | Monthly                                                                                                             | As per IS: 456                                                                                            | C                                                                                   | R/W      | R/W   |         |
| 8 CONCRETE TESTING EQUIPMENTS           |                                      |                                                                                                                          |                                               |                                                                                                                     |                                                                                                           |                                                                                     |          |       |         |
|                                         | Concrete Compression Testing Machine | IS: 516, IS:1199, IS 14858 Clause 10.1.1                                                                                 | As per Specifications                         | Calibration Report - Once in Year                                                                                   |                                                                                                           | R                                                                                   | R        | R     |         |
| 9 OTHER MEASURING DEVICES               |                                      |                                                                                                                          |                                               |                                                                                                                     |                                                                                                           |                                                                                     |          |       |         |
|                                         | Total Station                        | As per Manufacturer Spec                                                                                                 | As per valid guidelines                       | Calibration Report - Once in a year                                                                                 |                                                                                                           | R                                                                                   | R        | R     |         |
|                                         | Levelling Instrument                 | IS: 2988                                                                                                                 | As per valid guidelines                       | Calibration Report - Once in Six months                                                                             |                                                                                                           |                                                                                     |          |       |         |
|                                         | Weighing Machines                    | As per Manufacturer Spec                                                                                                 | As per valid guidelines                       | Calibration Report - Once in a Year                                                                                 |                                                                                                           |                                                                                     |          |       |         |
|                                         | Cube moulds & Slump Cone             | IS: 516, 1199                                                                                                            | As per valid guidelines                       | Calibration Report - Once in a Year                                                                                 |                                                                                                           |                                                                                     |          |       |         |
|                                         | Measuring jars                       | As per Manufacturer Spec                                                                                                 | As per valid guidelines                       | Calibration Report - Once in a Year                                                                                 |                                                                                                           |                                                                                     |          |       |         |
|                                         | Oven                                 | As per Manufacturer Spec                                                                                                 | As per valid guidelines                       | Calibration Report - Once in a Year                                                                                 |                                                                                                           |                                                                                     |          |       |         |
| 10 LIFTING DEVICES                      |                                      |                                                                                                                          |                                               |                                                                                                                     |                                                                                                           |                                                                                     |          |       |         |
|                                         | Tower Crane                          | As per Manufacturer Spec                                                                                                 | As per valid guidelines                       | Fitness Certificate - Once in a year                                                                                |                                                                                                           | R                                                                                   | R        | R     |         |
|                                         | Men / Material Hoist                 | As per Manufacturer Spec                                                                                                 | As per valid guidelines                       | Fitness Certificate - Once in a Year                                                                                |                                                                                                           |                                                                                     |          |       |         |
|                                         | Hydra Machine / Crane                | As per Manufacturer Spec                                                                                                 | As per valid guidelines                       | Fitness Certificate - Once in a Year                                                                                |                                                                                                           |                                                                                     |          |       |         |
| 11 WATER PROOFING MEMBRANES & CHEMICALS |                                      |                                                                                                                          |                                               |                                                                                                                     |                                                                                                           |                                                                                     |          |       |         |
|                                         | Visual Inspection                    | Purchase order                                                                                                           | IMIR                                          | Random                                                                                                              | Purchase order Specification                                                                              | C                                                                                   | R        | R     |         |
|                                         | Physical Properties                  | IS: 13435 & Manufacturer's Technical Data sheet                                                                          | MTC/<br>Third Party Lab Report                | MTC : Every Batch                                                                                                   | As per TDS                                                                                                | R                                                                                   | R        | R     |         |
|                                         | Chemical Properties                  |                                                                                                                          |                                               |                                                                                                                     |                                                                                                           |                                                                                     |          |       |         |
| 12 SOLID CONCRETE BLOCKS                |                                      |                                                                                                                          |                                               |                                                                                                                     |                                                                                                           |                                                                                     |          |       |         |
|                                         | Visual Inspection                    | Purchase order                                                                                                           | IMIR                                          | Random/ each consignment                                                                                            | Purchase order Specification                                                                              | C                                                                                   | R        | R     |         |
|                                         | Dimensions                           | IS: 2185 Clause 11 / BOQ/Technical Specifications                                                                        | Internal Lab Report                           | Internal Lab Report -Min. One set of sample Per 5000 Blocks Per source<br>Third party Lab : One per source approval | Variation:<br>In length = +/- 5 mm<br>In Height = +/-3mm                                                  | C                                                                                   | R        | R/C   |         |
|                                         | Compressive Strength                 |                                                                                                                          | Internal Lab Report<br>Third Party Lab Report |                                                                                                                     | Minimum 35 Kg/sq.cm as per BOQ                                                                            |                                                                                     |          |       |         |
|                                         | Density                              |                                                                                                                          | Internal Lab Report                           |                                                                                                                     | Min 1800 Kg/cum                                                                                           |                                                                                     |          |       |         |
|                                         | Water Absorption                     |                                                                                                                          | Internal Lab Report                           |                                                                                                                     | Max 10%                                                                                                   |                                                                                     |          |       |         |

|                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                | <b>BCIM Engineering Pvt. Ltd.</b><br>INSPECTION AND TEST PLAN |                                   |                                                                                                                             |                                                             |                |  |       |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|---------------------------------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------|-------|---------|---------------|--|-----------|--|--|--------------------------|-----------|--|--|--|--|--|------|--|--|------|--|--|--|--|--|-------------|--|--|-------------|--|--|--|
| PROJECT                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                | Lanco Hills Residential Project                               |                                   |                                                                                                                             |                                                             |                |                                                                                     |       |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
| BUILDINGS                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                | LH10                                                          |                                   |                                                                                                                             |                                                             |                |                                                                                     |       |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
| PART - A - MATERIAL & EQUIPMENT                                                                                                                                                                                                                                                                                                                                                                                                               |                                                |                                                               |                                   |                                                                                                                             |                                                             |                |                                                                                     |       |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
| Sl.No.                                                                                                                                                                                                                                                                                                                                                                                                                                        | Inspection and Testing Description             | Controlling Specification                                     | Verifying Document                | Frequency of Inspec./Testing                                                                                                | Acceptance Criteria                                         | Responsibility |                                                                                     |       | Remarks |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                |                                                               |                                   |                                                                                                                             |                                                             | Vendor         | Client                                                                              |       |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                |                                                               |                                   |                                                                                                                             |                                                             |                | Projects                                                                            | QA/QC |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
| 13                                                                                                                                                                                                                                                                                                                                                                                                                                            | PLASTER MESH                                   |                                                               |                                   |                                                                                                                             |                                                             |                |                                                                                     |       |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                               | Make and Type (Visual Inspection)              | As per Technical Spec                                         | IMIR                              | IMIR - Random/ each consignment<br>MTC - One per Source / Year<br>Third Party Lab Report - One per source / Year            | As per Manufacturer's Specification                         | R              | R                                                                                   | R     |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                               | Thickness                                      |                                                               | MTC                               |                                                                                                                             |                                                             | R              | R                                                                                   | R     |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                               | Mass (g/sq.m)                                  |                                                               | Third Party Lab Report            |                                                                                                                             |                                                             | R              | R                                                                                   | R     |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                               | Zinc coating                                   |                                                               | MTC                               |                                                                                                                             |                                                             | R              | R                                                                                   | R     |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                               | Mesh size                                      |                                                               | Third Party Lab Report            |                                                                                                                             |                                                             | R              | R                                                                                   | R     |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                               | Tensile Strength:<br>Length wise<br>Cross wise |                                                               | Third Party Lab Report            |                                                                                                                             |                                                             | R              | R                                                                                   | R     |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
| 14                                                                                                                                                                                                                                                                                                                                                                                                                                            | SAND FOR PLASTER MORTAR                        |                                                               |                                   |                                                                                                                             |                                                             |                |                                                                                     |       |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                               | Visual Inspection                              | Purchase order                                                | IMIR                              | Random/ each consignment                                                                                                    | Purchase order Specification                                | C              | R                                                                                   | R/C   |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                               | Grading Analysis of sand                       | Technical specifications / IS :1542                           | Internal / Third Party Lab Report | Internal Lab Report : One test per 250 Cubic meters per Source<br>Third Party Lab Report : One test per 6 months per source | Limits As per IS 1542                                       | C/R            | R                                                                                   | R/W   |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                               | Field silt content test                        |                                                               |                                   |                                                                                                                             |                                                             | C/R            | R                                                                                   | R/W   |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
| 15                                                                                                                                                                                                                                                                                                                                                                                                                                            | HIGH STRENGTH ANCHOR BOLTS                     |                                                               |                                   |                                                                                                                             |                                                             |                |                                                                                     |       |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                               | Visual Inspection                              | Purchase order                                                | IMIR                              | Random                                                                                                                      | Purchase order Specification                                | C              | R                                                                                   | R/C   |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                               | Pull Out Test                                  | MTC                                                           | Third Party Lab Report            | Third Party Lab Report : One test per source / size                                                                         | MTC & Structural Consultant Approval                        | R              | R                                                                                   | R/C   |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                               | Tensile Strength                               | MTC                                                           | MTC                               | MTC : Every Batch                                                                                                           | As per Technical Specification of BOQ / As per Approved MAR | R              | R                                                                                   | R/C   |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                               | Yield Strength                                 | MTC                                                           | MTC                               | MTC : Every Batch                                                                                                           | As per Technical Specification of BOQ / As per Approved MAR | R              | R                                                                                   | R/C   |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                               | Elongation %                                   | MTC                                                           | MTC                               | MTC : Every Batch                                                                                                           | As per Technical Specification of BOQ / As per Approved MAR | R              | R                                                                                   | R/C   |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
| 16                                                                                                                                                                                                                                                                                                                                                                                                                                            | EXPANSION JOINT FILLERS                        |                                                               |                                   |                                                                                                                             |                                                             |                |                                                                                     |       |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                               | Visual Inspection                              | Purchase order                                                | IMIR                              | Random/ each consignment                                                                                                    | Purchase order Specification                                | C              | R                                                                                   | R/C   |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
| <table border="1"> <tr> <td colspan="2">Approved By :</td> <td>Signature</td> <td colspan="2"></td> <td rowspan="3">Received &amp; Accepted By :</td> <td>Signature</td> <td colspan="3"></td> </tr> <tr> <td colspan="2"></td> <td>Name</td> <td colspan="2"></td> <td>Name</td> <td colspan="3"></td> </tr> <tr> <td colspan="2"></td> <td>Designation</td> <td colspan="2"></td> <td>Designation</td> <td colspan="3"></td> </tr> </table> |                                                |                                                               |                                   |                                                                                                                             |                                                             |                |                                                                                     |       |         | Approved By : |  | Signature |  |  | Received & Accepted By : | Signature |  |  |  |  |  | Name |  |  | Name |  |  |  |  |  | Designation |  |  | Designation |  |  |  |
| Approved By :                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                | Signature                                                     |                                   |                                                                                                                             | Received & Accepted By :                                    | Signature      |                                                                                     |       |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                | Name                                                          |                                   |                                                                                                                             |                                                             | Name           |                                                                                     |       |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                | Designation                                                   |                                   |                                                                                                                             |                                                             | Designation    |                                                                                     |       |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |
| LEGEND: W: Witness, R: Review, C: Check, MTC: Manufacturer's Test Certificate, IMIR: Incoming materials Inspection Report, MAR: Material Approval Report, TDS: Technical Data Sheet, OMC: Optimum Moisture Content, MS: Method Statement                                                                                                                                                                                                      |                                                |                                                               |                                   |                                                                                                                             |                                                             |                |                                                                                     |       |         |               |  |           |  |  |                          |           |  |  |  |  |  |      |  |  |      |  |  |  |  |  |             |  |  |             |  |  |  |